

# AI Friday Season 2 – Problem Statements

## Audit & Compliance Automated Policy Q&A Assistant

### PROBLEM STATEMENT

Audit and compliance teams often struggle to quickly interpret complex corporate policies and regulatory requirements when performing audits. Manual review of lengthy policy documents delays audit cycles and increases risk of non-compliance. Auditors require quick access to relevant policy interpretations and clarifications during fieldwork. Existing tools do not provide an interactive, conversational means to query policy documents contextually. This gap leads to inconsistent audit outcomes, prolonged audit durations, and increased compliance risks.

### Data Considerations:

Corporate policy documents and regulatory texts in PDF or DOCX format. Historical audit queries and resolution logs. API access to audit management platforms for integration. Data preprocessing to extract structured text and index documents for search. Privacy measures to protect sensitive policy content. Synthetic data generation by anonymizing policies for testing conversational capabilities.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

### SOLUTION EXPECTATIONS

A web-based conversational assistant that enables auditors to upload policy documents and ask natural language questions, receiving concise, context-aware answers. Features include follow-up query handling, keyword highlighting, and exportable Q&A transcripts. Integration with audit tools for seamless workflow. Powered by GenAI document understanding and conversational capabilities. Success metrics include question accuracy, reduction in review time, and user satisfaction. Demonstration through policy upload and live Q&A session.

Feel free to explore creative solutions that align with the problem context and objectives.

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## Banking Loan Eligibility Pre-Assessment Chatbot

### PROBLEM STATEMENT

Loan applicants often face uncertainty about their eligibility before applying, resulting in wasted time for both customers and bank staff. Manual pre-assessment processes are slow and inconsistent, leading to customer dissatisfaction and increased workload for loan officers. Current digital tools are limited to static eligibility criteria without personalized guidance. There is a need for an AI-powered chatbot that can interactively gather applicant data, perform preliminary eligibility assessments using bank policies, and provide instant feedback, improving customer experience and operational efficiency.

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### Data Considerations:

Requires access to credit scoring APIs, bank loan policy rules, and applicant data inputs. Data quality and privacy compliance are critical, including adherence to financial regulations. Synthetic applicant data can be generated for testing. Preprocessing involves data validation and normalization.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

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### SOLUTION EXPECTATIONS

A conversational AI interface that collects applicant data, validates inputs, and generates instant loan eligibility feedback with explanations. Features include integration with credit APIs and policy databases, multi-turn dialogue, and fallback handling. The generative AI interprets eligibility criteria and produces user-friendly responses. Success metrics include 90% eligibility assessment accuracy and reduced pre-screening time. Demonstrations include live applicant interactions and backend validations. Documentation covers usage and integration.

Feel free to explore creative solutions that align with the problem context and objectives.

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## IT Application Security Compliance Checker

### PROBLEM STATEMENT

Application development teams often struggle to ensure that their code complies with the organization's security policies and regulatory standards. Manual code reviews for security compliance are time-consuming and prone to oversight, increasing vulnerability risks. Current static analysis tools lack contextual understanding of policy requirements and generate many false positives. There is a need for an AI-powered assistant that analyzes code snippets for security compliance, provides explanations, and suggests remediation steps in an interactive interface, streamlining compliance and reducing risks.

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### Data Considerations:

Security policies in structured format (e.g., JSON schemas). Sample code repositories with annotated compliance issues. Access to CI/CD tools API. Data preprocessing to parse and tokenize code snippets. Privacy and IP protection for proprietary code. Synthetic data through code snippet generation with known compliance violations.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

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### SOLUTION EXPECTATIONS

A web-based tool where developers submit code snippets for AI-powered analysis against security policies. Features include violation detection, explanation, remediation suggestions, and integration hooks for CI/CD systems. Powered by GenAI code understanding and policy mapping. Success measured by detection accuracy, developer adoption, and reduced security incidents. Demonstration includes code submission and detailed compliance feedback.

Feel free to explore creative solutions that align with the problem context and objectives.

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## AI-Powered Automated Code Review Comment Generator

### PROBLEM STATEMENT

Code reviews are critical for software quality but require significant manual effort from developers. Reviewing code for style, logic errors, and best practices can be time-consuming and inconsistent. Existing static analysis tools provide limited contextual feedback and often overwhelm developers with false positives. An AI-based code review assistant that generates context-aware, constructive review comments can improve code quality and accelerate development cycles.

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### Data Considerations:

Requires access to Python code samples and corresponding review comments from open-source repositories for prompt engineering. Data volume is flexible. Data quality must ensure relevance and accuracy. Privacy is minimal. Synthetic code-review pairs may be generated for testing. Preprocessing includes code formatting and tokenization.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

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### SOLUTION EXPECTATIONS

An interactive web application where developers submit code snippets and receive AI-generated review comments highlighting issues and suggestions. Features include comment editing, code highlighting, and integration with repository APIs. The generative AI uses language models fine-tuned on code-review data. Success metrics include 85% comment relevance and reduced review times. Demonstrations include live code review and feedback generation. Documentation covers usage and integration.

Feel free to explore creative solutions that align with the problem context and objectives.

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## Capital Markets Trading Strategy Idea Generator with Generative AI

### PROBLEM STATEMENT

Traders and analysts in capital markets constantly seek innovative trading strategies but face challenges in brainstorming and validating ideas quickly. Manual research and analysis are time-intensive and may miss emerging trends. Existing tools provide historical data and analytics but lack creative assistance. There is a need for an AI-driven assistant that generates novel trading strategy ideas based on market data and user input, fostering innovation and competitive advantage.

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### Data Considerations:

Requires access to real-time and historical market data via APIs (e.g., Bloomberg, Alpha Vantage). Data formats include time-series and structured financial indicators. Data volume is large; efficient filtering is necessary. Data quality and freshness are critical. Synthetic market scenarios can be created for development. Preprocessing includes normalization and feature extraction.

Consider using synthetic or anonymized data where appropriate. Ensure data quality and relevance to the problem context.

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### SOLUTION EXPECTATIONS

An AI-powered interactive platform where users input trading preferences and receive AI-generated strategy ideas with rationale and market context. Features include idea ranking, detailed descriptions, and integration with market data APIs. The generative AI synthesizes data and user inputs to create novel concepts. Success metrics include 80% expert-rated relevance and increased ideation speed. Demonstrations showcase idea generation and exploration. Documentation covers usage and integration.

Feel free to explore creative solutions that align with the problem context and objectives.